

Zara E-Commerce Management System

Order and Order_Details Analysis

1.Database Design

1.1Order table

Attribute	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Order_Date	DATE	NOT NULL
Total_Amount	NUMBER(10,2)	NOT NULL

1.2 Order_Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Product_ID	INT	Foreign Key
Quantity	INT	NOT NULL
Unit_Price	NUMBER(10,2)	NOT NULL
Subtotal	NUMBER(10,2)	NOT NULL

2.Table Relationships

Order_Details connects each order with the corresponding Product using Order_ID and Product_ID.

ORDERS

|

| 1 : M

| One order can have many order details

↓

ORDER_DETAILS

|

| M : 1

| Each order detail belongs to one product

↓

PRODUCT

3. Table Creation

3.1 Create Order table

```
SQL> CREATE TABLE Orders (  
2     Order_ID INT PRIMARY KEY,  
3     Customer_ID INT,  
4     Order_Date DATE NOT NULL,  
5     Total_Amount NUMBER(10,2) NOT NULL,  
6     FOREIGN KEY (Customer_ID) REFERENCES Customer(CUSTOMERID)  
7 );
```

Table created.

3.2 Create Order_details

```
SQL> CREATE TABLE Order_Details (  
2     Order_Detail_ID INT PRIMARY KEY,  
3     Order_ID INT,  
4     Product_ID INT,  
5     Quantity INT NOT NULL,  
6     Unit_Price NUMBER(10,2) NOT NULL,  
7     Subtotal NUMBER(10,2) NOT NULL,  
8     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),  
9     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
10 );
```

Table created.

4. Order Data insertion

```

SQL> INSERT INTO Orders VALUES
  2 (1001, 108, TO_DATE('20-09-2026','DD-MM-YYYY'), 2500.00);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1002, 224, TO_DATE('21-09-2026','DD-MM-YYYY'), 1800.00);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1003, 567, TO_DATE('22-09-2026','DD-MM-YYYY'), 3200.00);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1004, 108, TO_DATE('23-09-2026','DD-MM-YYYY'), 1500.00);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1005, 224, TO_DATE('24-09-2026','DD-MM-YYYY'), 4200.00);

1 row created.

```

5. Order_Details data insertion

```

SQL> INSERT INTO Order_Details VALUES
  2 (1, 1001, 101, 1, 2500.00, 2500.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details VALUES
  2 (2, 1002, 102, 2, 900.00, 1800.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details VALUES
  2 (3, 1003, 103, 2, 1600.00, 3200.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details VALUES
  2 (4, 1004, 104, 1, 1500.00, 1500.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details VALUES
  2 (5, 1005, 105, 2, 2100.00, 4200.00);

1 row created.

```

6. Order Modification operations

6.1 Modify order amount

```
SQL> UPDATE Orders
  2 SET Total_Amount = 2700.00
  3 WHERE Order_ID = 1001;

1 row updated.
```

```
SQL> SELECT * FROM Orders;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
-----
      1001         108 20-SEP-26          2700
      1002         224 21-SEP-26          1800
```

6.2 Modify Order date

```
SQL> UPDATE Orders
  2 SET Order_Date = TO_DATE('25-09-2026', 'DD-MM-YYYY')
  3 WHERE Order_ID = 1001;

1 row updated.
```

7. Order_Details Modification

```
SQL> UPDATE Order_Details
  2 SET Quantity = 3,
  3     Subtotal = 2700.00
  4 WHERE Order_Detail_ID = 2;

1 row updated.
```

8. Customer Order history report

```
SQL> SELECT
  2   O.Order_ID,
  3   O.Customer_ID,
  4   O.Order_Date,
  5   O.Total_Amount,
  6   OD.Product_ID,
  7   OD.Quantity,
  8   OD.Unit_Price,
  9   OD.Subtotal
 10 FROM Orders O
 11 JOIN Order_Details OD
 12 ON O.Order_ID = OD.Order_ID
 13 ORDER BY O.Order_Date;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	PRODUCT_ID	QUANTITY	UNIT_PRICE
SUBTOTAL						
1002	224	21-SEP-26	1800	102	3	900
2700						
1003	567	22-SEP-26	3200	103	2	1600
3200						